



EYES IN THE SKY

AI BIRD DETECTION FOR DRONE SAFETY

“PI”RONMAN

AGENDA & OUR TEAM

Our Team

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Live Demo

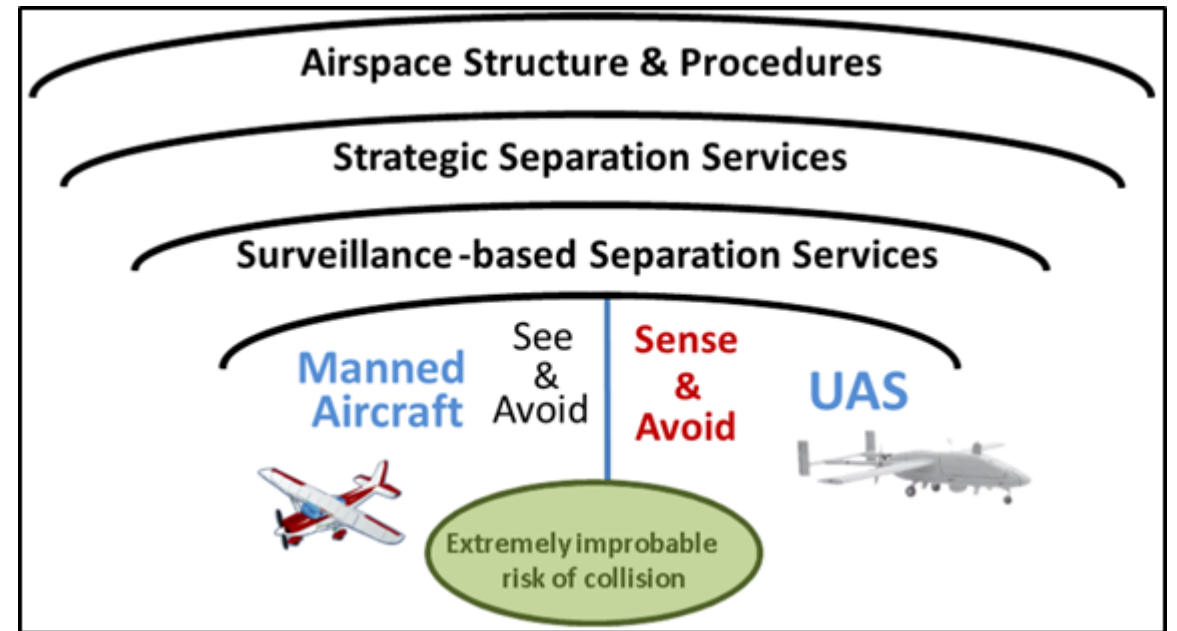
Name	Role	Description
Jordan Capelle	Systems Integration Engineer	Hardware setup & YOLO integration
Karl Garman	Research & Analysis Engineer	Literature review & documentation
Mohomad Sabur	Lead Software Engineer	AI model coding & optimization

MOTIVATION

Goal: Implement real-time bird detection on Raspberry Pi 5 and compare performance with and without Hailo-8L.

Why: Birds are unpredictable collision hazards for sUAS; onboard detection could support collision avoidance.

- This is a collision hazard that the FAA's airspace management framework does not consider.



LITERATURE GAP

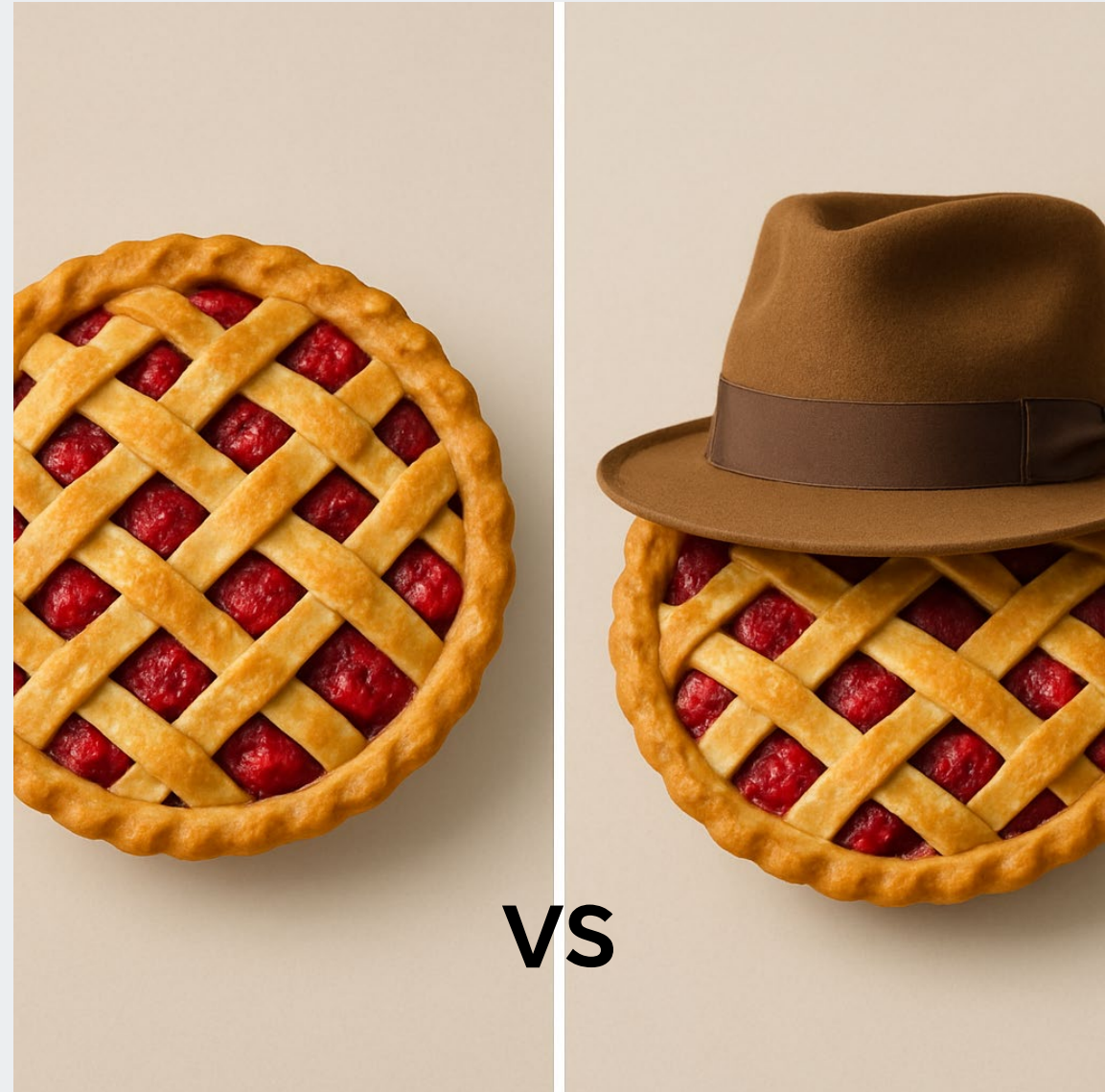
None of the reviewed papers used a YOLO model on an AI chip and applied it to the use case of identifying avian flight hazards.

→ This presented an opportunity

- ALQAYSI, ET.AL., "A TEMPORAL BOOSTED YOLO-BASED MODEL FOR BIRDS DETECTION AROUND WIND FARMS", JOURNAL OF IMAGING, 7, 227, 2021.
- ARRIOLA-VALVERDE, ET.AL., "COFFEE CROP DETECTION FROM UAS ORTHOMAPS WITH CONVOLUTIONAL NEURAL NETWORKS", 2023 IEEE CONFERENCE ON AGRIFOOD ELECTRONICS, 2023.
- DI CECIO, ET.AL., "ON-BOARD DRONE CLASSIFICATION WITH DEEP LEARNING AND SYSTEM-ON-CHIP IMPLEMENTATION", J. PHYS.: CONF. SER. **2716** 012059, 2024.
- GALLAGHER, ET.AL., "SURVEYING YOU ONLY LOOK ONCE (YOLO) MULTISPECTRAL OBJECT DETECTION ADVANCEMENTS, APPLICATIONS, AND CHALLENGES", IEEE ACCESS, DOI 10.1109/ACCESS.2025.3526458, 2024.
- GERARDI, "THE APPLICABILITY OF MACHINE LEARNING METHODS ON INFRARED VIDEO DATA", U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND ARMAMENTS CENTER, 2020.
- LAWRENCE, ET.AL., "UAS-BASED REAL-TIME DETECTION OF RED-COCKADED WOODPECKER CAVITIES IN HETEROGENEOUS LANDSCAPES USING YOLO OBJECT DETECTION ALGORITHMS", REMOTE SENS. 15, 883, 2023.
- SCHERMANN, ET.AL., "RISK-AWARE INTRUSION DETECTION AND PREVENTION SYSTEM FOR AUTOMATED UAS", 2023 IEEE 34TH ISSREW, 2023.
- SUN, ET.AL., "FBD-SV-2024: FLYING BIRD OBJECT DETECTION DATASET IN SURVEILLANCE VIDEO", [WWW.NATURE.COM/SCIENTIFICDATA/](https://www.nature.com/scientificdata/), 12:530 | [HTTPS://DOI.ORG/10.1038/S41597-025-04872-6](https://doi.org/10.1038/S41597-025-04872-6), 2025.
- TIN, ET.AL., "EXPLORING AN ALTERNATIVE APPROACH: RCNN-SVM FOR WEED IDENTIFICATION IN AERIAL IMAGES CAPTURED BY UAS", 2024 INTERNATIONAL CONFERENCE ON EMERGING TRENDS IN NETWORKS AND COMPUTER COMMUNICATIONS, 2024.
- ULLAH, ET.AL., "AUTONOMOUS CONTROL WITH VISION AND DEEP LEARNING: A RASPBERRY PI EDGE COMPUTING PLATFORM FOR OBSTACLE DETECTION IN SUAV PATH", 7TH INTERNATIONAL COMMNET CONFERENCE, 2024.

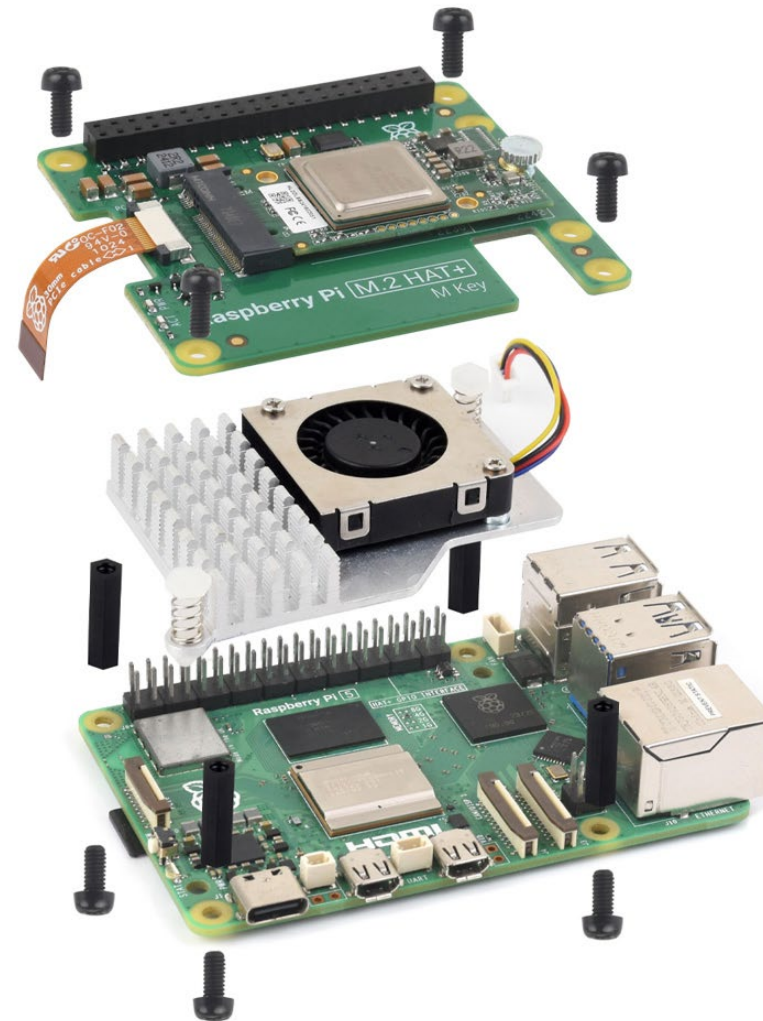
APPROACH OVERVIEW

- Platform comparison: Raspberry Pi 5 vs Raspberry Pi + Hailo-8L
- Object detection using YOLO
- Bird-only class filtering
- Benchmarking method: stills, video, robustness, stability



HARDWARE

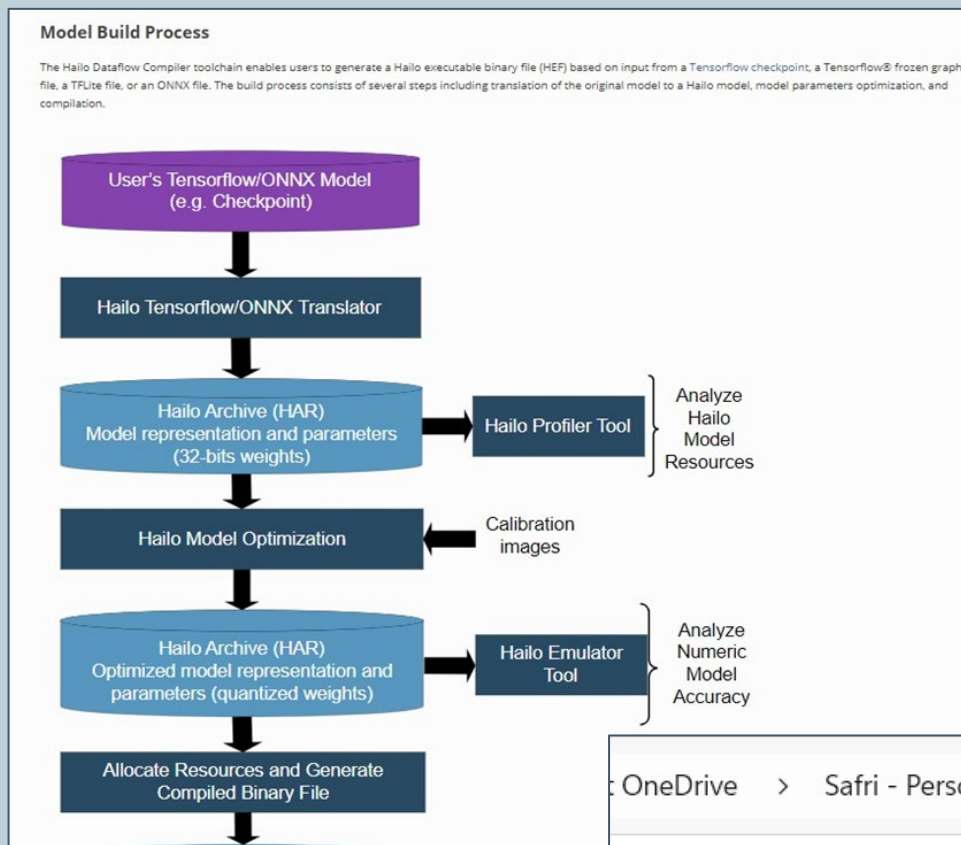
- Raspberry Pi 5 baseline
- Hailo-8L AI HAT (13TOPS) PCIe Gen3
- Low power, small form factor



AI MODEL INTEGRATION: PROBLEM

- Hailo 8L – Uses Proprietary .hef
- YOLO Model is .pt
- Hailo Dataflow Compiler cannot run on Pi5

Hailo DFC Flow



Conversion Flow



OneDrive > Safri - Personal > Desktop > UVA Classes

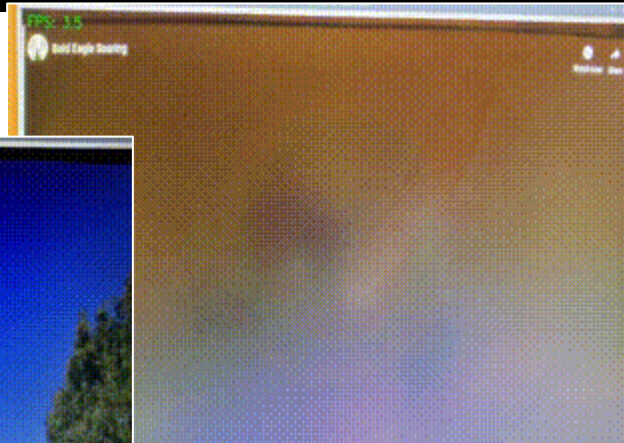
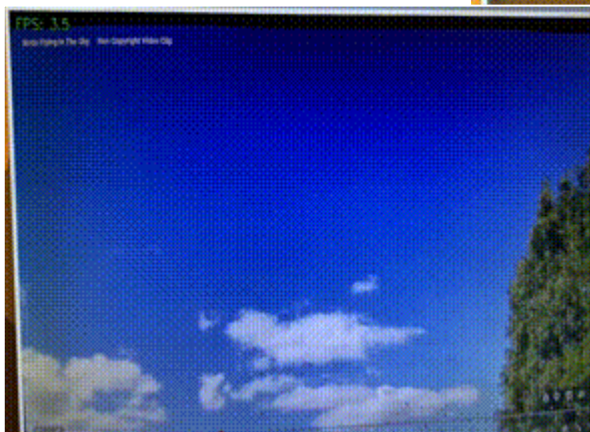
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yolov11n.har	11/18/2025 7:26 AM	HA
yolov11n.hef	11/18/2025 7:30 AM	HEI
yolov11n.onnx	11/18/2025 7:28 AM	ON
yolov11n_quantized.har	11/18/2025 7:32 AM	HA

HARDWARE VALIDATION

PI5 CPU ONLY = 6.25 FPS

```
safri@raspberrypi: ~  
File Edit Tabs Help  
safri@raspberrypi:~ $ source ~/yolo_cpu_venv/bin/activate  
(yolo_cpu_venv) safri@raspberrypi:~ $ python3 yolo_cpu_benchmark.py  
2025-11-16 15:28:09.675766783 [W:onnxruntime:Default, device_discovery.cc  
U device discovery failed: device_discovery.cc:89 ReadFileContents Failed  
1/device/vendor"  
Loading YOLOv8n ONNX model on CPU...  
Input name: images  
Input shape: [1, 3, 640, 640]  
YOLOv8n CPU-only FPS on Raspberry Pi 5: 6.25 FPS  
(yolo_cpu_venv) safri@raspberrypi:~ $
```

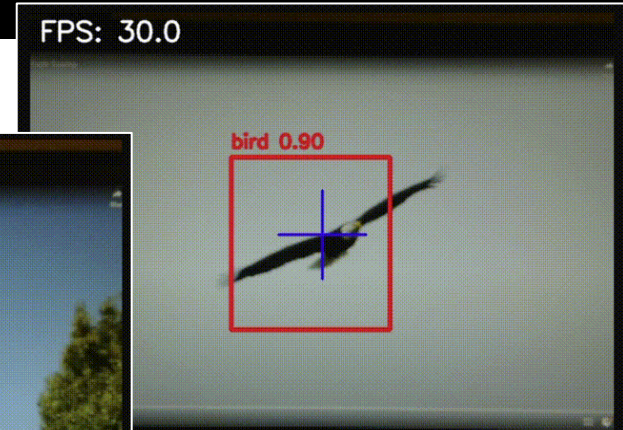
FPS: 3.5



PI5 + AI HAT = 58.67 FPS

```
safri@raspberrypi: ~  
File Edit Tabs Help  
safri@raspberrypi:~ $ hailortcli benchmark ~/Downloads/yolov8s_h8l.hef  
Starting Measurements...  
Measuring FPS in HW-only mode  
Network yolov8s/yolov8s: 100% | 882 | FPS: 58.67 | ETA: 00:00:00  
Measuring FPS (and Power on supported platforms) in streaming mode  
Network yolov8s/yolov8s: 100% | 882 | FPS: 58.67 | ETA: 00:00:00  
Measuring HW Latency  
Network yolov8s/yolov8s: 100% | 881 | HW Latency: 12.92 ms | ETA: 00:00:00  
=====  
Summary  
=====  
FPS (hw_only) = 58.6732  
FPS (streaming) = 58.6738  
Latency (hw) = 12.9192 ms  
safri@raspberrypi:~ $
```

FPS: 30.0



EXPERIMENTAL TEST SETUP

- Raspberry Pi 5 with camera
- Camera aimed at external monitor
- Slideshow of bird and non-bird images/videos
- Fixed distance and indoor lighting
- Same setup used for all tests

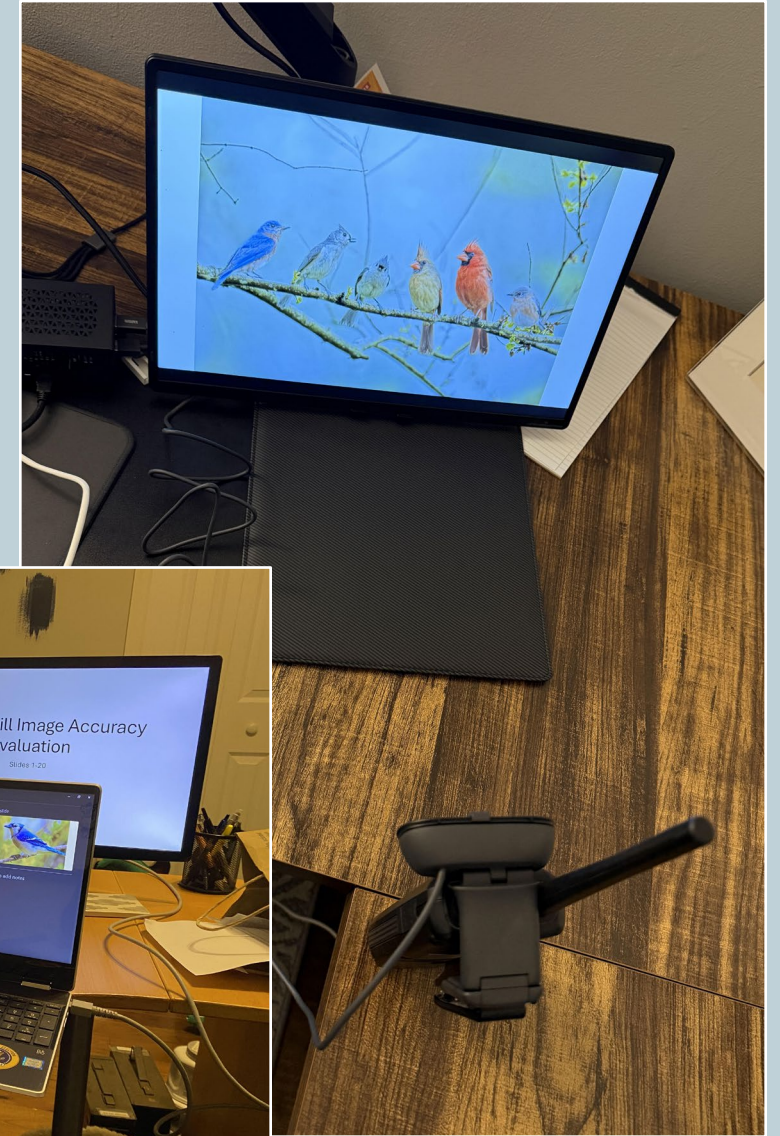
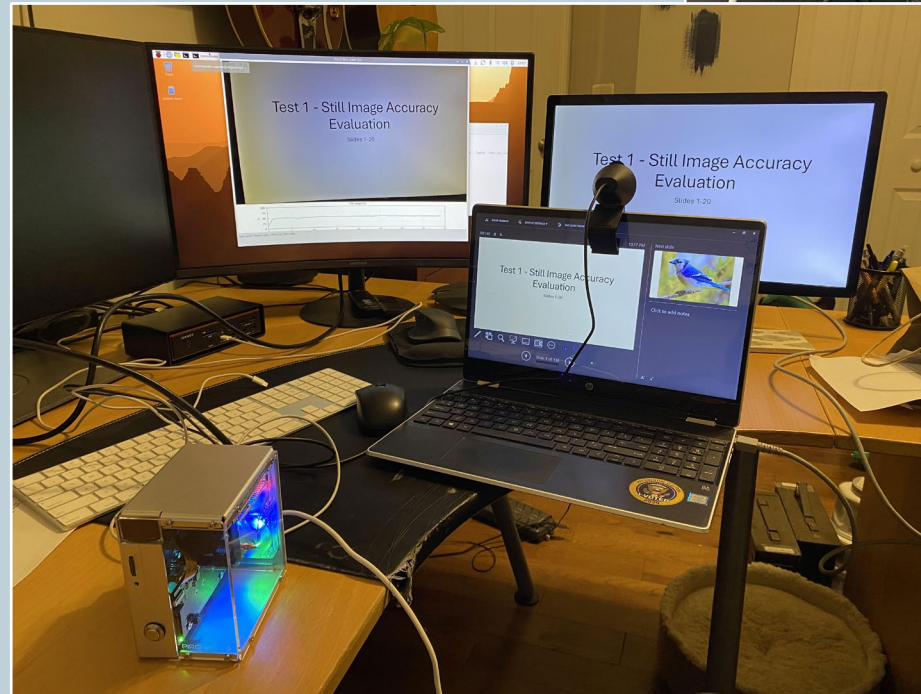


Image Examples:



TEST PROCEDURE

Four Tests

- 1. Still image classification
- 2. Video detection
- 3. Robustness (challenging images)
- 4. Stability (heat + long run)

Performance Metrics

- FPS
- Stability (crashes, slowdowns)
- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
 - True Positive (TP)
 - True Negative (TN)
 - False Positive (FP)
 - False Negative (FN)

Table 2 — Still Image Accuracy Summary

Tester Name	Platform	Total Samples	TP	FN	FP	TN	Precision	Recall	Notes
Jordan	Pi5	20	9	1	0	10	1	0.9	Failed to detect the partially occluded

RESULTS

Still images

- Pi only: ~0.9 precision, one missed detection
- Pi+HAT: perfect recall, occasional kite FP

Video

- Pi only: ~3–4 FPS
- Pi+HAT: 30 FPS

Robustness

- Pi misses occluded + small birds
- Pi+HAT recovers them

Stability

- No crashes; HAT runs hotter

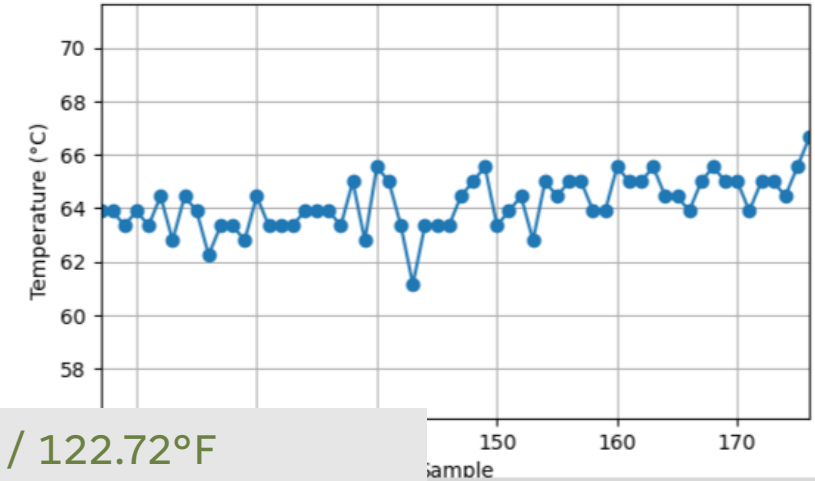
With Hat →

CPU Temperature

66.7 °C / 152.0 °F

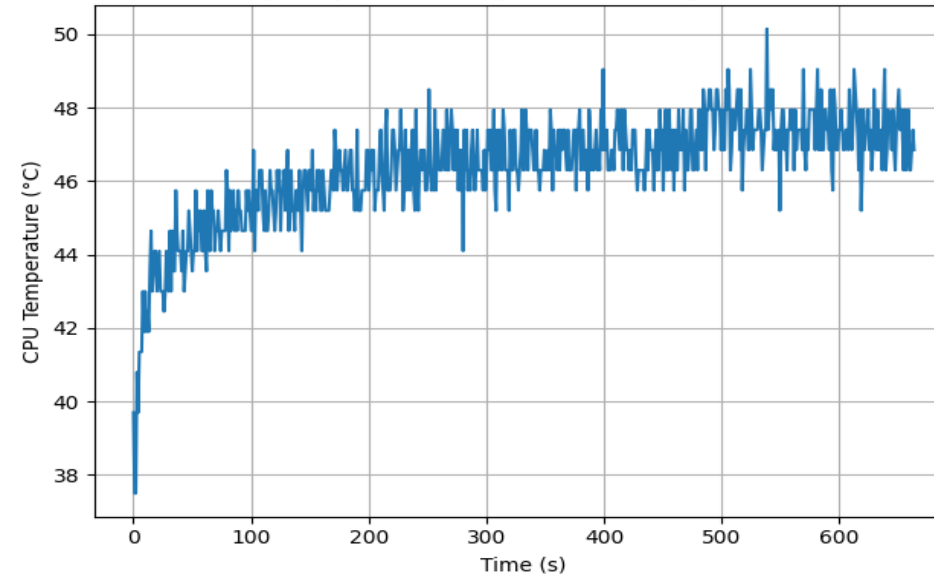
Last updated: 19:54:33 (every 10 s)

CPU Temperature Over Time



CPU Temperature: 50.4°C / 122.72°F

Raspberry Pi CPU Temperature During Inference



INTERPRETATION:

CPU-only is not sufficient for collision-avoidance timing



Pi5 functions, but struggles with real-time performance



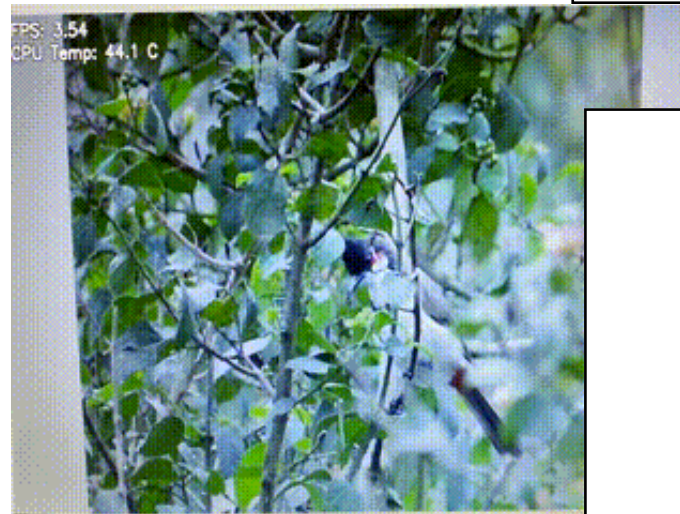
Hailo significantly improves latency and FPS



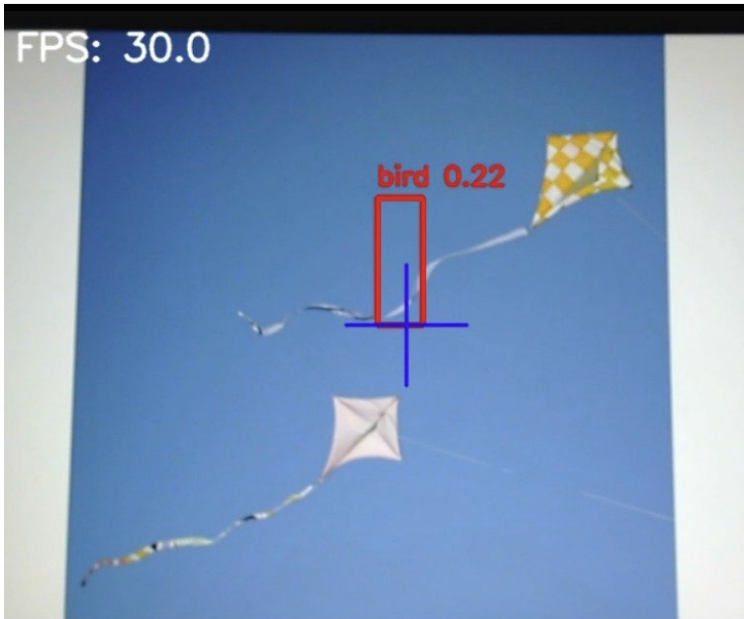
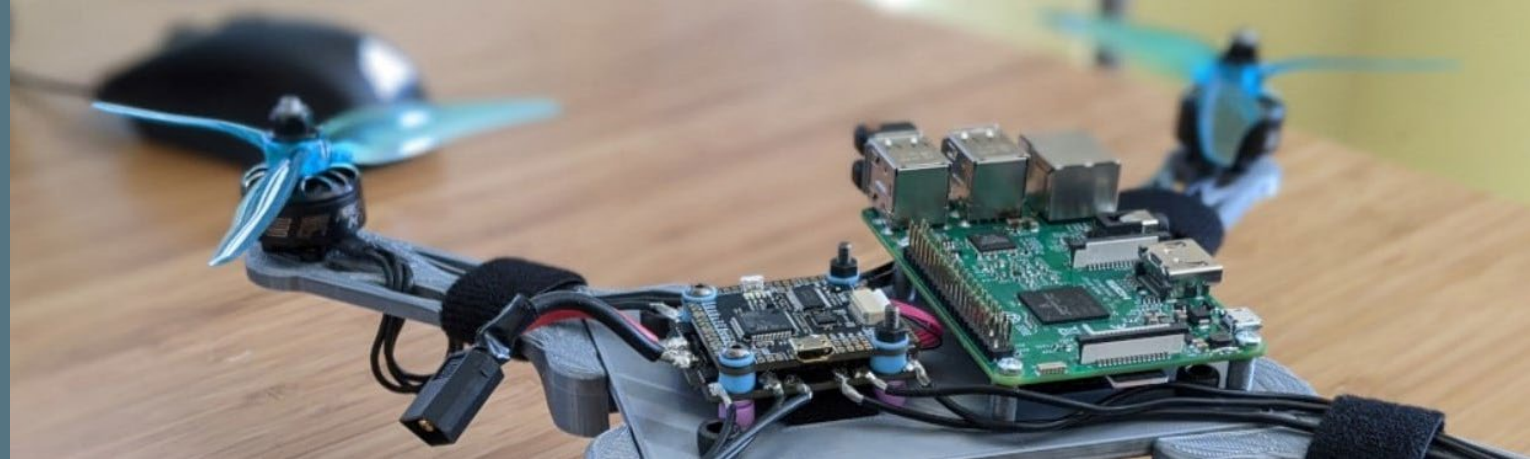
Accuracy and detection reliability improved under challenging conditions



System appears deployable for future UAV tests



LIMITATIONS AND RECOMMENDATIONS FOR FUTURE WORK



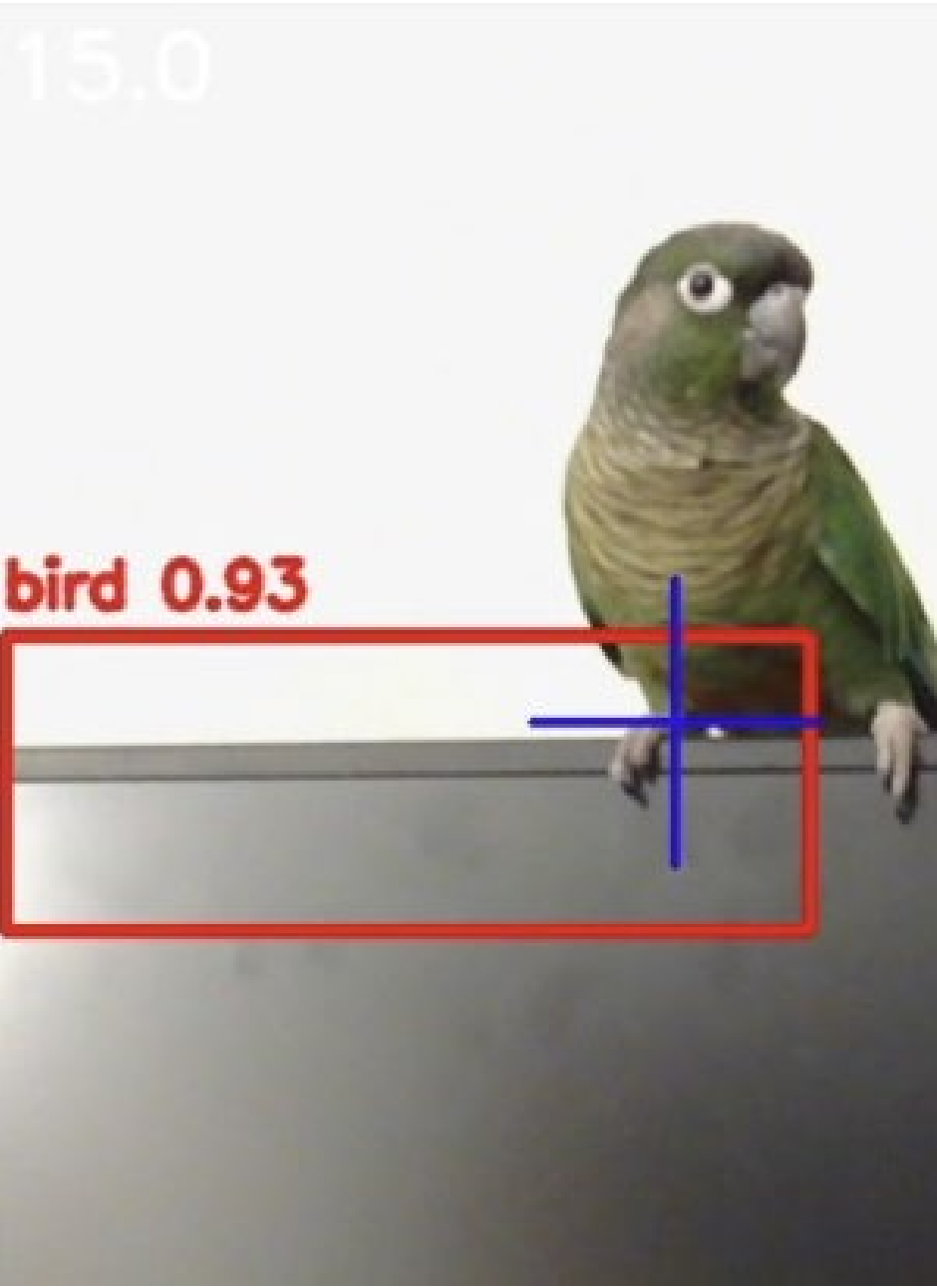
Limitations

- Controlled indoor test environment (bench test only, not real birds)
- YOLO11n pretrained, minimal tuning
- Remaining false positives (kite)



Recommendations

- Evaluate actual flight tests
- Try 26-TOPS HAT
- Include more airborne hazards
- Measure end-to-end latency more precisely



CONCLUSION

- Hailo-8L clearly improves performance
- Real-time detection is achievable
- Demonstrated meaningful improvement in recall, FPS, and responsiveness
- The Pi alone can detect birds but is too slow for real-time avoidance
- The Hailo AI HAT provides real-time inference capability
- Edge AI appears viable for sUAS collision avoidance
- These findings help to address a gap in literature and pave the way for AI research sUAS



THANK YOU

Jordan Capelle, Karl Garman, Mohomad Sabur